# Transcorrelated Jastrow Fitting: Atomization Energies

## 1 Theory

The selected virtual space is constructed by projecting the large-basis virtual MOs onto the small-basis AO space via

$$M = S_{11}^{-1/2} S_{12} C^{\text{virt}} \qquad [n_{\text{small}} \times n_{\text{large,virt}}] \tag{1}$$

where $S_{11}$ and $S_{12}$ are the small-basis and cross-overlap matrices, and $C^{\text{virt}}$ collects the large-basis virtual MO coefficients. The elements $M_{\mu a} = \langle \tilde{\varphi}_\mu | \psi_a \rangle$ are overlaps between Löwdin-orthonormalized small AOs and $S_{22}$-orthonormal large-basis virtuals, so the thin SVD

$$M = U \Sigma V^\top \qquad U\ [n_{\text{small}} \times n_{\text{small}}],\ \Sigma\ [n_{\text{small}} \times n_{\text{small}}],\ V\ [n_{\text{large,virt}} \times n_{\text{small}}] \tag{2}$$

yields singular values $\sigma_k \leq 1$ that are exact cosines of the principal angles between the two subspaces. Dropping the $n_{\text{occ}}$ columns of $V$ with near-zero $\sigma_k$ and projecting onto the AO basis gives the $n_{\text{sel}} = n_{\text{small}} - n_{\text{occ}}$ selected virtuals

$$C^{\text{sel}} = C^{\text{virt}} V_{\text{kept}} \qquad [n_{\text{large,AO}} \times n_{\text{sel}}] \tag{3}$$

which are $S_{22}$-orthonormal by construction. The SVD rotation mixes canonical virtuals, rendering the projected Fock matrix

$$F^{\text{sel}} = V_{\text{kept}}^\top \operatorname{diag}(\varepsilon_a)\, V_{\text{kept}} \qquad [n_{\text{sel}} \times n_{\text{sel}}] \tag{4}$$

non-diagonal. Diagonalizing $F^{\text{sel}} = W \tilde{\Lambda} W^\top$ yields the semicanonical orbitals

$$C^{\text{final}} = C^{\text{virt}}\, V_{\text{kept}}\, W \qquad [n_{\text{large,AO}} \times n_{\text{sel}}]. \tag{5}$$

## 2 Atomization energies

We compute atomization energies with several xTC method families and benchmark them against SHCI+PBE+CV reference values. All calculations employ effective core potentials (ECPs) for the third-row elements. The SHCI, SHCI+PBE, and conventional CCSD(T) results used here as references are taken from Ref. [2]; the SHCI+PBE energies extrapolated to the complete basis set (CBS) limit are considered the best available benchmark for the G2/55 set. Per-basis, per-molecule method discrepancies and the full basis set convergence of error metrics are hosted as interactive plotly figures [1].

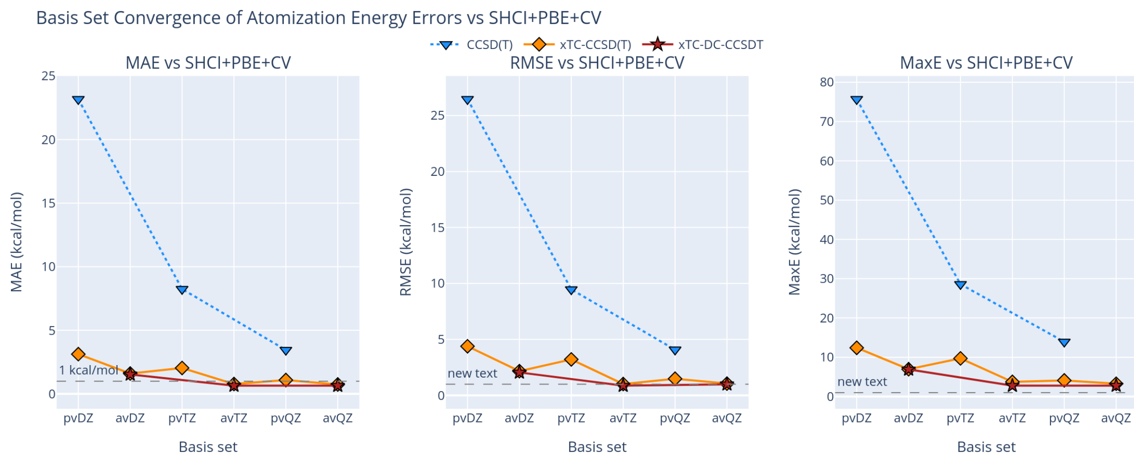


Figure 1: Basis set convergence (MAE, RMSE, MaxE in kcal/mol) of atomization energy errors vs SHCI+PBE+CV for each method family, on a unified x-axis interleaving cc-pV$x$Z and aug-cc-pV$x$Z. The interactive version and per-basis per-molecule breakdowns are available online [1].

Table 1: MAE, RMSE, and MaxE of atomization energies vs SHCI+PBE+CV reference (kcal/mol), for each method and basis set.

| Method | Basis | MAE | RMSE (kcal/mol) | MaxE |
|---|---|---|---|---|
| *SHCI* | | | | |
| | CBS | 0.18 | 0.26 | 0.75 |
| *xTC-CCSD(T)* | | | | |
| | pvDZ | 3.13 | 4.38 | 12.40 |
| | avDZ | 1.60 | 2.16 | 6.95 |
| | pvTZ | 2.04 | 3.21 | 9.70 |
| | avTZ | 0.78 | 1.01 | 3.73 |
| | pvQZ | 1.09 | 1.48 | 4.10 |
| | avQZ | 0.74 | 1.05 | 3.26 |
| *xTC-DC-CCSDT* | | | | |
| | avDZ | 1.53 | 2.05 | 6.90 |
| | avTZ | 0.65 | 0.87 | 2.77 |
| | avQZ | 0.65 | 0.99 | 2.78 |
| *CCSD(T)* | | | | |
| | pvDZ | 23.18 | 26.47 | 75.71 |
| | pvTZ | 8.25 | 9.46 | 28.60 |
| | pvQZ | 3.45 | 4.08 | 13.94 |
| | pv5Z | 1.67 | 1.96 | 5.52 |

# References

[1] Kristoffer Simula. Interactive plotly figures: xtc atomization energies vs shci+pbe+cv. Online collection, GitHub Pages, 2026. `https://simulak1.github.io/tc-g2-data/`.

[2] Yuan Yao, Emmanuel Giner, Junhao Li, Julien Toulouse, and C. J. Umrigar. Almost exact energies for the Gaussian-2 set with the semistochastic heat-bath configuration interaction method. *J. Chem. Phys.*, 153(12):124117, 2020. doi:10.1063/5.0018577.